Retail Analysis with Walmart Data

The aim of this project is enable category managers of Walmart to check the weekly and monthly sales of the departments. Analysis includes the effect of markdowns on the sales and the extent of effect on the sales by fuel prices, temperature, unemployment, CPI etc. It Has been analyzed using simple linear regression models.

Business problem

The decision makers of Walmart should be able to analyze the effect of various factors affecting the sales of the products in their stores.

Analytics Problem

In this problem, we have analyzed sales across different departments by store type and created weekly and monthly. We have analyzed the effect of various factors such as temperature, store size, fuel prices, markdown in prices, unemployment and CPI to determine Which factors have a statistical significance in explaining sales in the stores by using simple linear regression.

Steps followed to carry out this task

- Imported the Dataset
- Perform Basic data cleaning
- Perform EDA
- Perform Statistical Analysis
- Model Building
- Model Evaluation

Basic Statistics tasks solution

Q1. Which store has maximum sales?

A: Store - 20 has a maximum sales = $301,397,792

Q2. Which store has maximum standard deviation i.e., the sales vary a lot. Also, find out the coefficient of mean to standard deviation

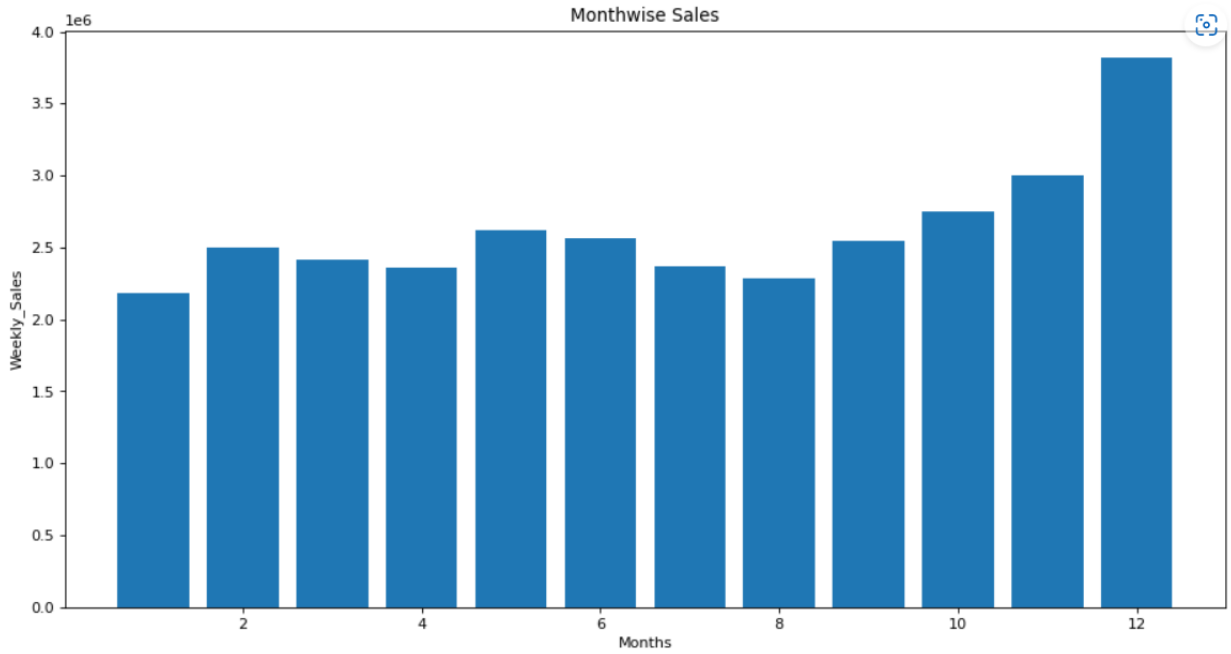
Store - 14 has a maximum standard deviation = $317,569.949
Mean to Standard Deviation = 15.71%

Q3. Which store/s has good quarterly growth rate in Q3’2012 ?
Store Number 20 has Good Quartely Growth in Q3'2012 301397792.46

Q4. Some holidays have a negative impact on sales. Find out holidays which have higher sales than the mean sales in non-holiday season for all stores together.

Thanksgiving has the highest sales (1,471,273.43) than non-holiday sales (1,041,256.38).

Q5. Provide a monthly and semester view of sales in units and give insights.



Statistical Model Result

Linear Regression

Train Accuracy Score: 12.754999999999999%
Test Accuracy Score: 12.504000000000001%
Mean Absolute Error: 421588.638
Mean Squared Error: 260734361065.515
Root Mean Squared Error: 510621.544

Random Forest Regressor *

Accuracy = 90.586 %
Mean Absolute Error = 94579.82507
Mean Squared Error = 28052252979.79217
Root Mean Squared Error = 167488.06817

Decision Tree Regressor *

Accuracy = 56.56 %
Mean Absolute Error = 156948.05351
Mean Squared Error = 129448103555.59921
Root Mean Squared Error = 359788.97086

KNearest Neighbors *

Accuracy = 37.899 %
Mean Absolute Error = 325483.21979
Mean Squared Error = 185057788123.77875
Root Mean Squared Error = 430183.43544

Conclusion:

Here, we have used 4 different algorithms to know which model to use to predict the weekly sales. Linear Regression is not an appropriate model to use as accuracy is very low. However, Random Forest Regression gives accuracy of almost 90% . so, it is the best model to forecast weekly sales.